

Traffic Collision Data Analysis

1. Data Preparation

```
sqlite_db_abs_path = os.path.abspath("ETL Project 1/dataset/switrs.sqlite")
jdbc_url = f"jdbc:sqlite:{sqlite_db_abs_path}"

# Use a subquery in dbtable with an alias
dbtable_query = "(SELECT CAST(case_id AS INTEGER) AS case_id_num, * FROM collisions) AS subq"

df_collisions_full = spark.read \
    .format("jdbc") \
    .option("url", jdbc_url) \
    .option("driver", "org.sqlite.JDBC") \
    .option("dbtable", dbtable_query) \
    .option("partitionColumn", "case_id_num") \
    .option("lowerBound", 1) \
    .option("upperBound", 10000000) \
    .option("numPartitions", 24) \
    .load()
```

Total records ingested into PySpark: 9,424,334

root

```
|-- case_id_num: decimal(38,18) (nullable = true)
|-- case_id: string (nullable = true)
|-- jurisdiction: long (nullable = true)
|-- officer_id: string (nullable = true)
|-- reporting_district: string (nullable = true)
|-- chp_shift: string (nullable = true)
|-- population: string (nullable = true)
|-- county_city_location: string (nullable = true)
|-- county_location: string (nullable = true)
|-- special_condition: string (nullable = true)
|-- beat_type: string (nullable = true)
|-- chp_beat_type: string (nullable = true)
|-- city_division_lapd: string (nullable = true)
|-- chp_beat_class: string (nullable = true)
|-- beat_number: string (nullable = true)
|-- primary_road: string (nullable = true)
|-- secondary_road: string (nullable = true)
|-- distance: double (nullable = true)
|-- direction: string (nullable = true)
```

|-- intersection: long (nullable = true)
|-- weather_1: string (nullable = true)
|-- weather_2: string (nullable = true)
|-- state_highway_indicator: long (nullable = true)
|-- caltrans_county: string (nullable = true)
|-- caltrans_district: long (nullable = true)
|-- state_route: long (nullable = true)
|-- route_suffix: string (nullable = true)
|-- postmile_prefix: string (nullable = true)
|-- postmile: double (nullable = true)
|-- location_type: string (nullable = true)
|-- ramp_intersection: string (nullable = true)
|-- side_of_highway: string (nullable = true)
|-- tow_away: long (nullable = true)
|-- collision_severity: string (nullable = true)
|-- killed_victims: long (nullable = true)
|-- injured_victims: long (nullable = true)
|-- party_count: long (nullable = true)
|-- primary_collision_factor: string (nullable = true)
|-- pcf_violation_code: string (nullable = true)
|-- pcf_violation_category: string (nullable = true)
|-- pcf_violation: long (nullable = true)
|-- pcf_violation_subsection: string (nullable = true)
|-- hit_and_run: string (nullable = true)
|-- type_of_collision: string (nullable = true)
|-- motor_vehicle_involved_with: string (nullable = true)
|-- pedestrian_action: string (nullable = true)
|-- road_surface: string (nullable = true)
|-- road_condition_1: string (nullable = true)
|-- road_condition_2: string (nullable = true)
|-- lighting: string (nullable = true)
|-- control_device: string (nullable = true)
|-- chp_road_type: string (nullable = true)
|-- pedestrian_collision: long (nullable = true)
|-- bicycle_collision: long (nullable = true)
|-- motorcycle_collision: long (nullable = true)
|-- truck_collision: long (nullable = true)
|-- not_private_property: long (nullable = true)

```
|-- alcohol_involved: long (nullable = true)
|-- statewide_vehicle_type_at_fault: string (nullable = true)
|-- chp_vehicle_type_at_fault: string (nullable = true)
|-- severe_injury_count: long (nullable = true)
|-- other_visible_injury_count: long (nullable = true)
|-- complaint_of_pain_injury_count: long (nullable = true)
|-- pedestrian_killed_count: long (nullable = true)
|-- pedestrian_injured_count: long (nullable = true)
|-- bicyclist_killed_count: long (nullable = true)
|-- bicyclist_injured_count: long (nullable = true)
|-- motorcyclist_killed_count: long (nullable = true)
|-- motorcyclist_injured_count: long (nullable = true)
|-- primary_ramp: string (nullable = true)
|-- secondary_ramp: string (nullable = true)
|-- latitude: double (nullable = true)
|-- longitude: double (nullable = true)
|-- collision_date: string (nullable = true)
|-- collision_time: string (nullable = true)
|-- process_date: string (nullable = true)
```

2. Data Cleaning

2.1 Missing Values

	column_name	missing_count	missing_percentage
0	road_condition_2	9385372	99.586581
1	secondary_ramp	9333416	99.035285
2	primary_ramp	9256350	98.217550
3	route_suffix	9228103	97.917826
4	weather_2	9161454	97.210625
5	pcf_violation_code	8942638	94.888806
6	city_division_lapd	8821318	93.601500
7	postmile_prefix	8508289	90.280003
8	alcohol_involved	8479870	89.978454
9	ramp_intersection	8404347	89.177092
10	side_of_highway	6867317	72.867929
11	location_type	6867301	72.867759
12	caltrans_county	6867299	72.867738
13	latitude	6730338	71.414468
14	longitude	6730338	71.414468
15	state_route	6533326	69.324007
16	postmile	6533326	69.324007
17	caltrans_district	6533325	69.323997
18	pcf_violation_subsection	6046560	64.159016
19	reporting_district	5572350	59.127255

2.2 Fixing Columns

Rows before deduplication: 9213214
Rows after deduplication: 9213211
Duplicate records removed: 3

2.3 Outlier Analysis

	column	q1	q3	iqr	lower_bound	upper_bound	outlier_count	outlier_pct
0	distance	0.0	460.0	460.0	-690.0	1150.0	1053031	11.429577
1	killed_victims	0.0	0.0	0.0	0.0	0.0	64435	0.699376
2	injured_victims	0.0	1.0	1.0	-1.5	2.5	333939	3.624567
3	party_count	2.0	2.0	0.0	2.0	2.0	2922997	31.726148
4	severe_injury_count	0.0	0.0	0.0	0.0	0.0	228565	2.480840
5	motorcyclist_injured_count	0.0	0.0	0.0	0.0	0.0	211266	2.293077

Total records before validation: 9213211

Total records after validation: 9210618

Non-physical anomalies removed: 2593

3. Exploratory Data Analysis

3.1.1. Data Preparation

Q: Classify variables into categorical and numerical.

```
Numerical Variables (15):  
- distance  
- killed_victims  
- injured_victims  
- party_count  
- severe_injury_count  
- other_visible_injury_count  
- complaint_of_pain_injury_count  
- pedestrian_killed_count  
- pedestrian_injured_count  
- bicyclist_killed_count  
- bicyclist_injured_count  
- motorcyclist_killed_count  
- motorcyclist_injured_count  
- latitude  
- longitude
```

- Categorical Variables (44):
- case_id (encoded/flag)
 - jurisdiction (encoded/flag)
 - officer_id
 - chp_shift
 - population
 - county_city_location (encoded/flag)
 - county_location
 - special_condition (encoded/flag)
 - beat_type
 - chp_beat_type
 - chp_beat_class
 - beat_number
 - primary_road
 - secondary_road
 - direction
 - intersection (encoded/flag)
 - weather_1
 - state_highway_indicator (encoded/flag)
 - route_suffix
 - tow_away (encoded/flag)
 - collision_severity
 - primary_collision_factor
 - pcf_violation_category
 - pcf_violation (encoded/flag)
 - hit_and_run
 - type_of_collision
 - motor_vehicle_involved_with
 - pedestrian_action
 - road_surface
 - road_condition_1
 - lighting
 - control_device
 - chp_road_type (encoded/flag)
 - pedestrian_collision (encoded/flag)
 - bicycle_collision (encoded/flag)
 - motorcycle_collision (encoded/flag)
 - truck_collision (encoded/flag)
 - not_private_property (encoded/flag)
 - statewide_vehicle_type_at_fault
 - chp_vehicle_type_at_fault
 - primary_ramp
 - collision_date
 - collision_time
 - process_date

General purpose buckets (1) [Info](#)

Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

<

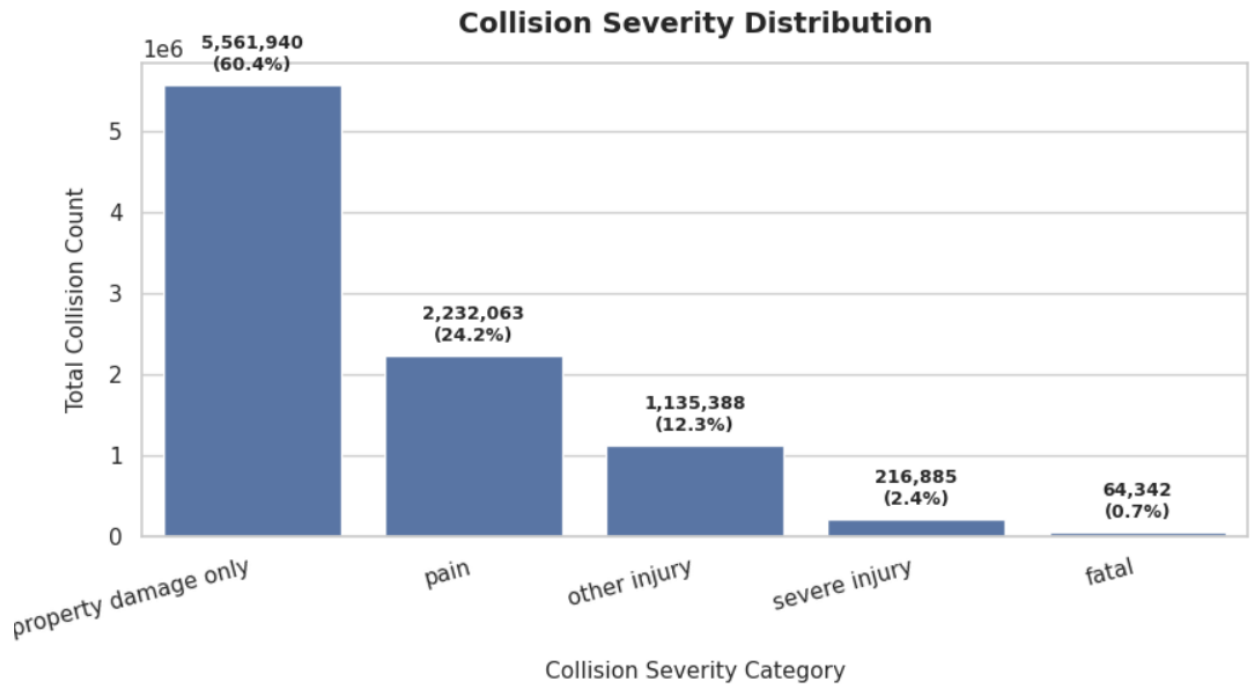
1

>

	Name ▲	AWS Region ▼	Creation date ▼
☐	s3-collision-etl-181040156891-us-east-1-an	US East (N. Virginia) us-east-1	September 7, 2026, 22:03:23 (UTC+05:30)

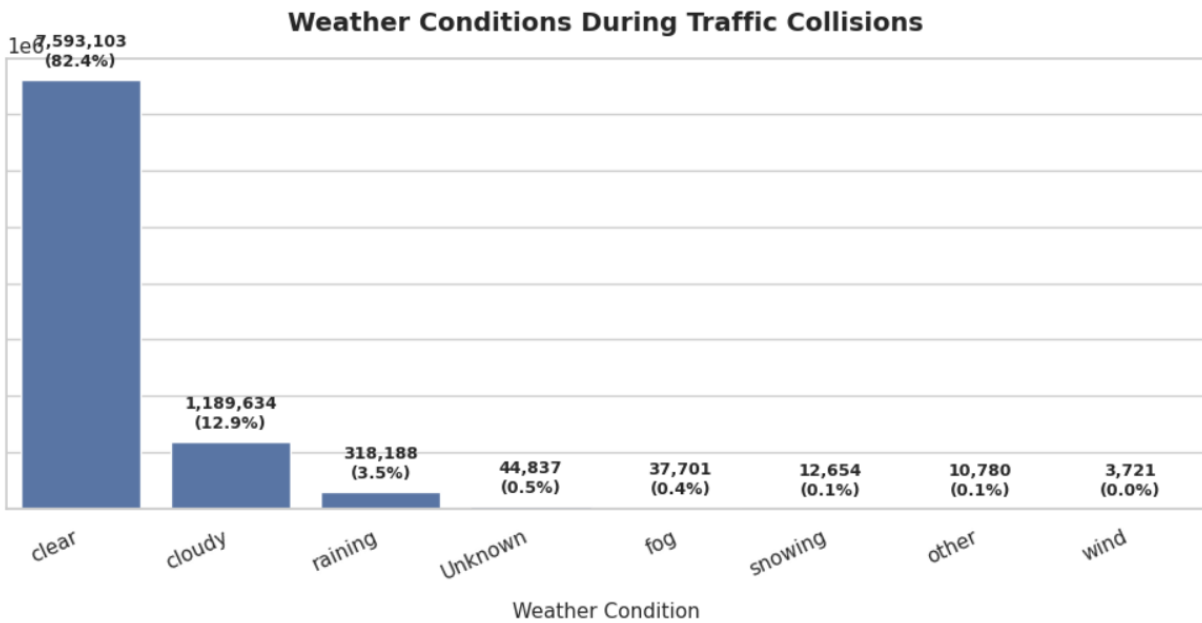
3.1.2. Analyze the distribution of collision severity

Q: Analyze the distribution of collision severity.



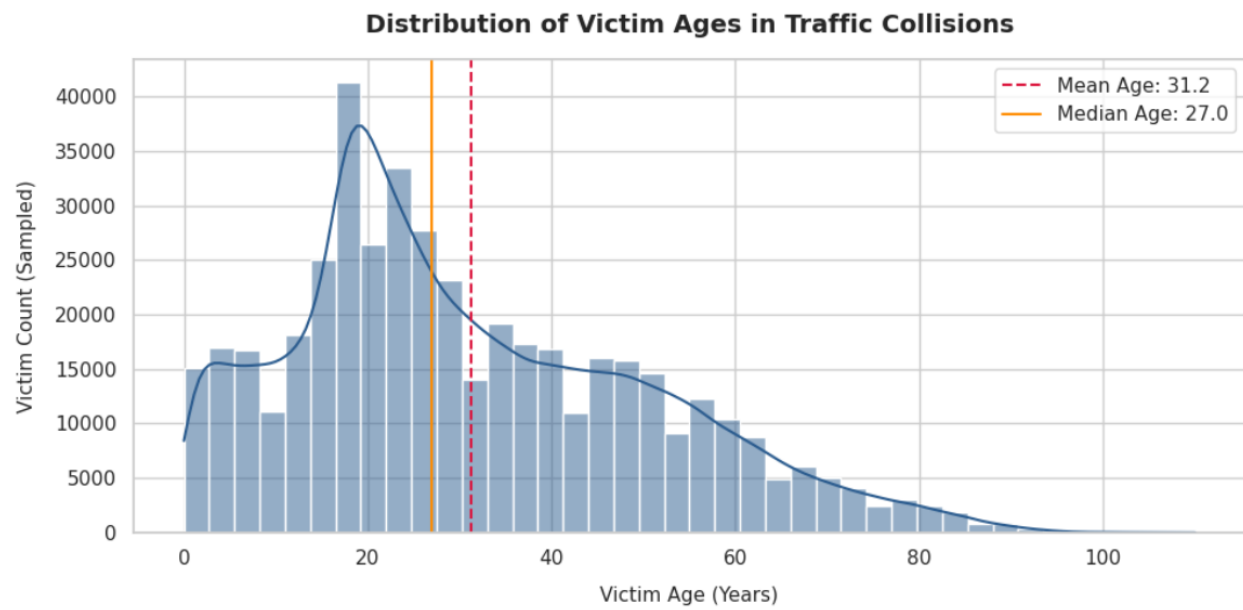
3.1.3. Weather conditions during collisions

Q: Examine weather conditions during collisions.



3.1.4. Victim Age Distribution

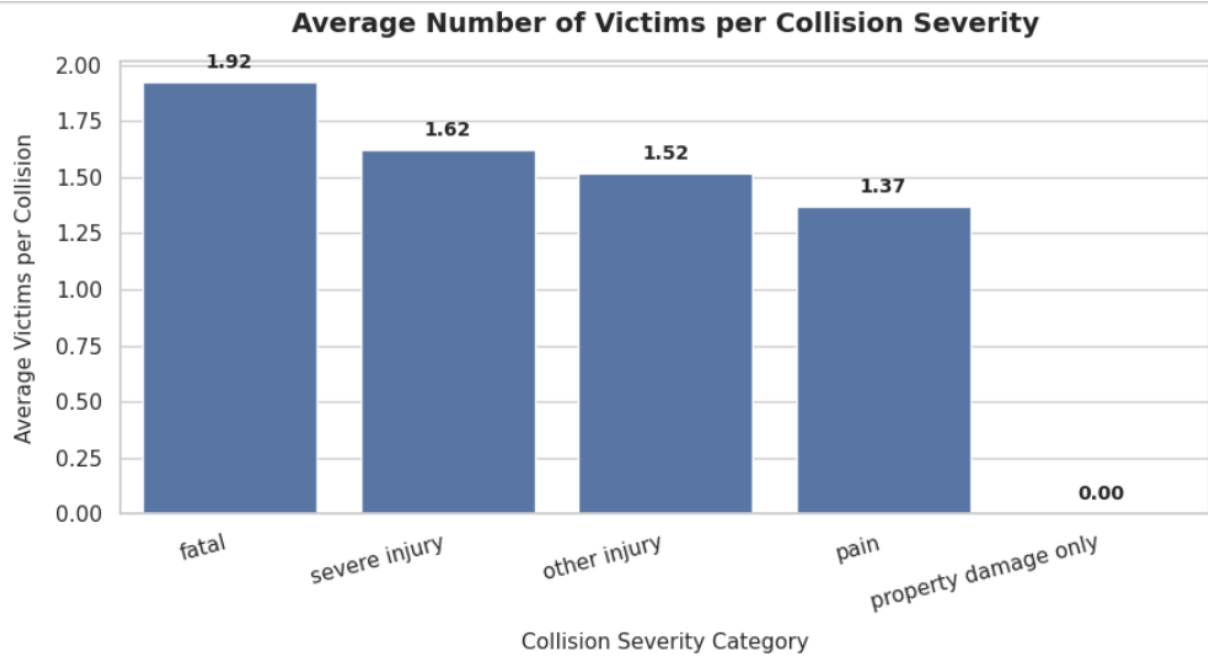
Q: Analyze the distribution of victim ages.



	count	unique	top	freq
victim_age	452379	110	18.000000000000000000	14661

3.1.5. Collision Severity vs Number of Victims.

Q: Study the relationship between collision severity and the number of victims.

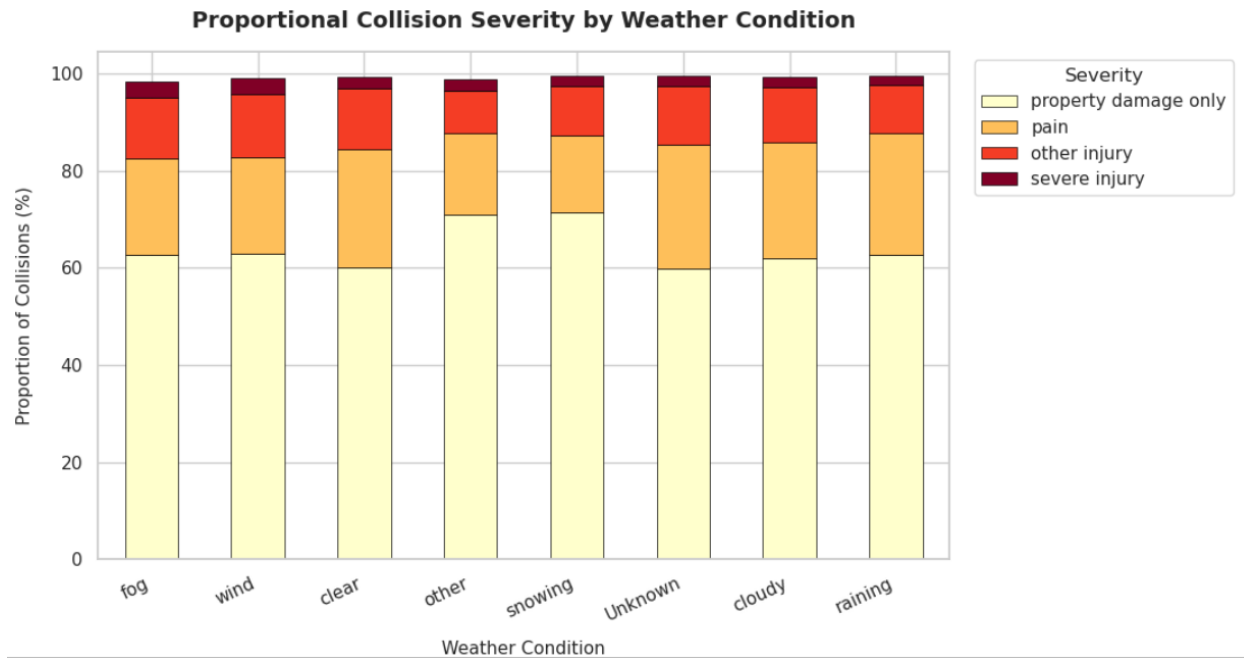


	severity	total_collisions	avg_victims_per_collision	avg_injured_victims	avg_killed_victims	total_victims_count
0	fatal	64342	1.922228	0.824516	1.097712	123680
1	severe injury	216885	1.622348	1.622348	0.000000	351863
2	other injury	1135388	1.516663	1.516663	0.000000	1722001
3	pain	2232063	1.368892	1.368892	0.000000	3055453
4	property damage only	5561940	0.000000	0.000000	0.000000	0

3.1.6. Weather Conditions vs Collision Severity

Q: Analyze the correlation between weather conditions and collision severity.

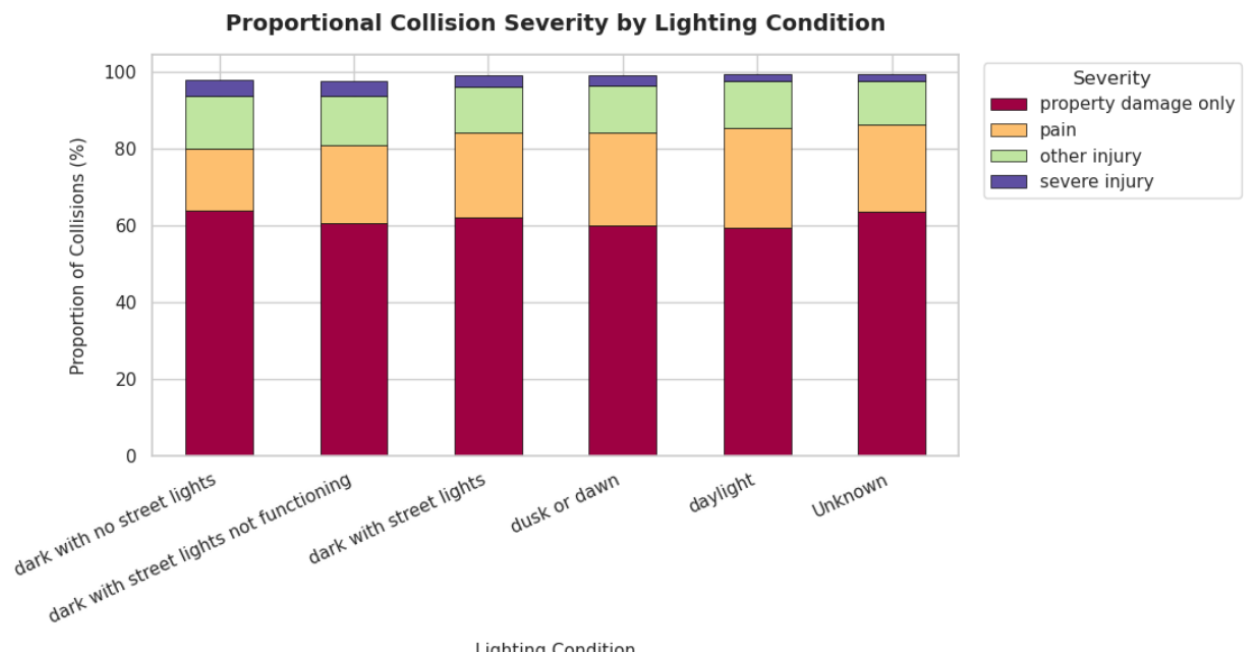
	severity	property damage only	pain	other injury	severe injury
weather_condition					
fog		62.70	19.71	12.73	3.32
wind		62.89	19.94	12.98	3.17
clear		59.99	24.32	12.58	2.41
other		71.00	16.60	8.82	2.38
snowing		71.46	15.88	10.05	2.16
Unknown		59.85	25.44	12.08	2.13
cloudy		61.98	23.77	11.42	2.09
raining		62.79	24.87	9.99	1.85



3.1.7. Lighting conditions vs Collision Severity.

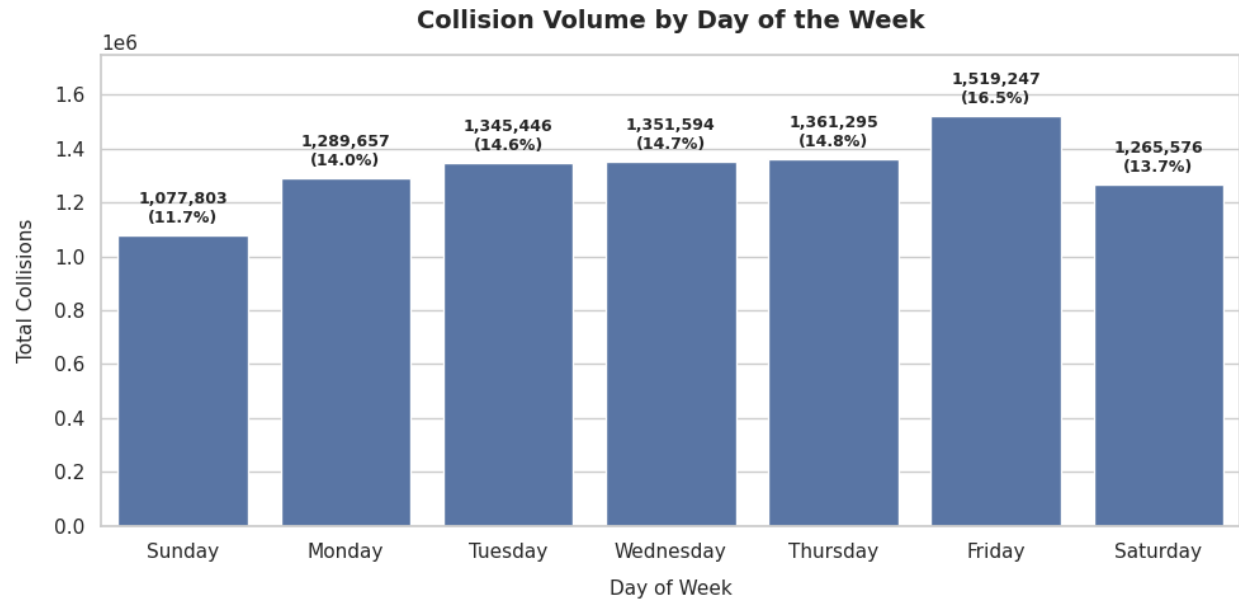
Q: Visualize the impact of lighting conditions on collision severity.

	severity	property damage only	pain	other injury	severe injury
lighting_condition					
dark with no street lights		64.02	16.03	13.68	4.25
dark with street lights not functioning		60.71	20.20	12.71	3.97
dark with street lights		62.20	21.92	12.02	2.93
dusk or dawn		59.98	24.04	12.37	2.71
daylight		59.37	25.98	12.27	1.93
Unknown		63.52	22.84	11.26	1.89



3.1.8. Weekday-Wise Collision Trends

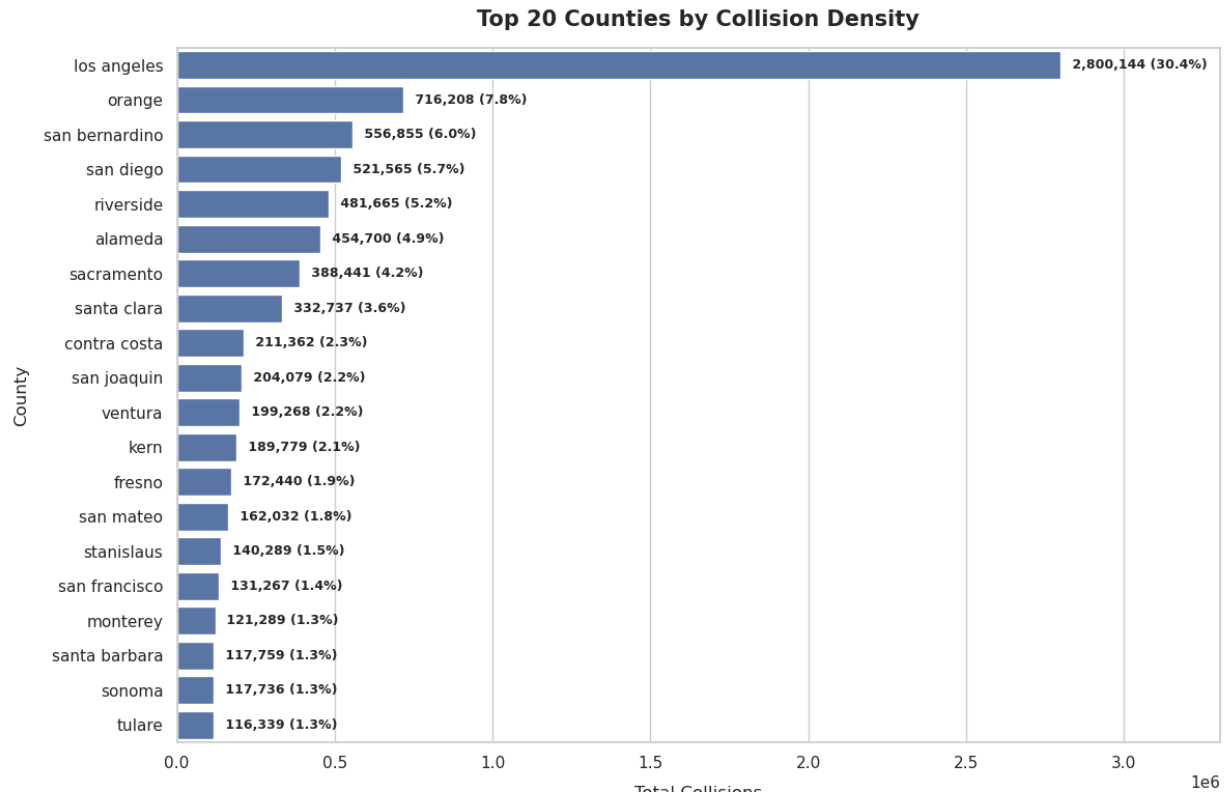
Q: Extract and analyze weekday-wise collision trends.



day_num	weekday	collision_count	percentage
0	1 Sunday	1077803	11.701745
1	2 Monday	1289657	14.001851
2	3 Tuesday	1345446	14.607554
3	4 Wednesday	1351594	14.674303
4	5 Thursday	1361295	14.779627
5	6 Friday	1519247	16.494518
6	7 Saturday	1265576	13.740403

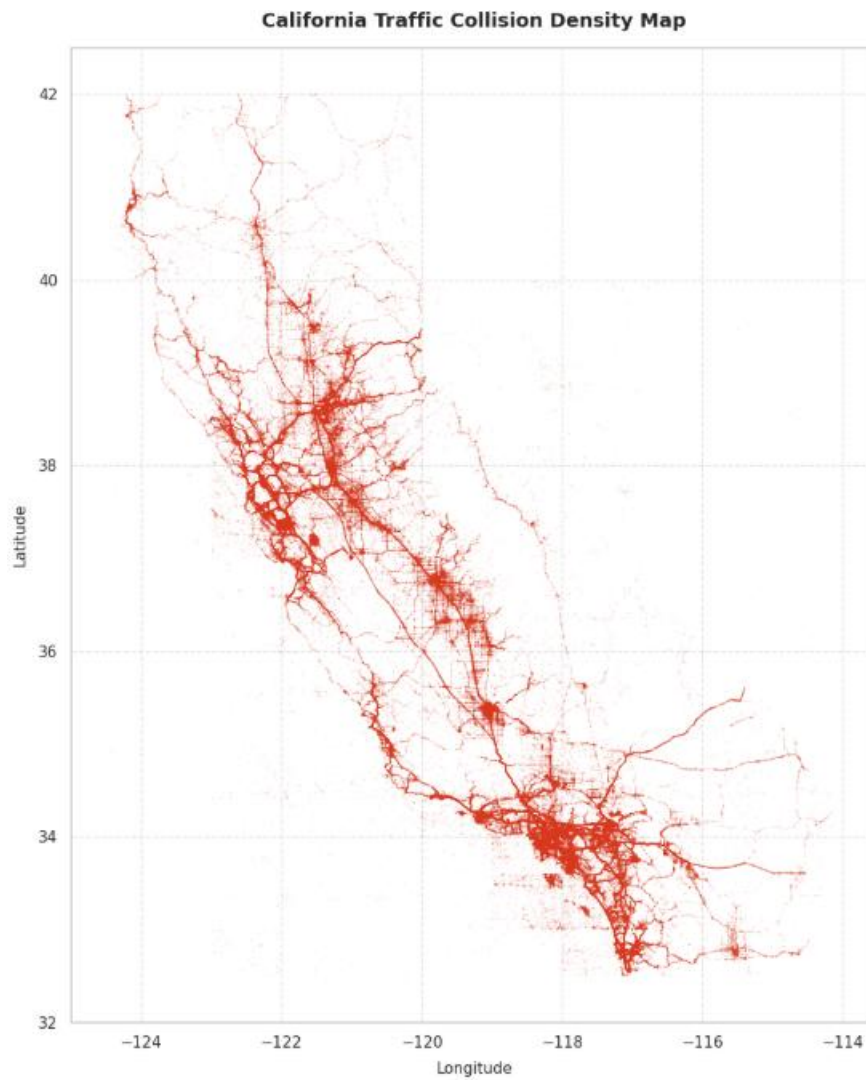
3.1.9. Spatial Distribution of Collisions

Q: Study spatial distribution of collisions by county.



3.1.10. Collision Analysis by Geography

Q: Generate a scatter plot to analyze collision locations geographically.

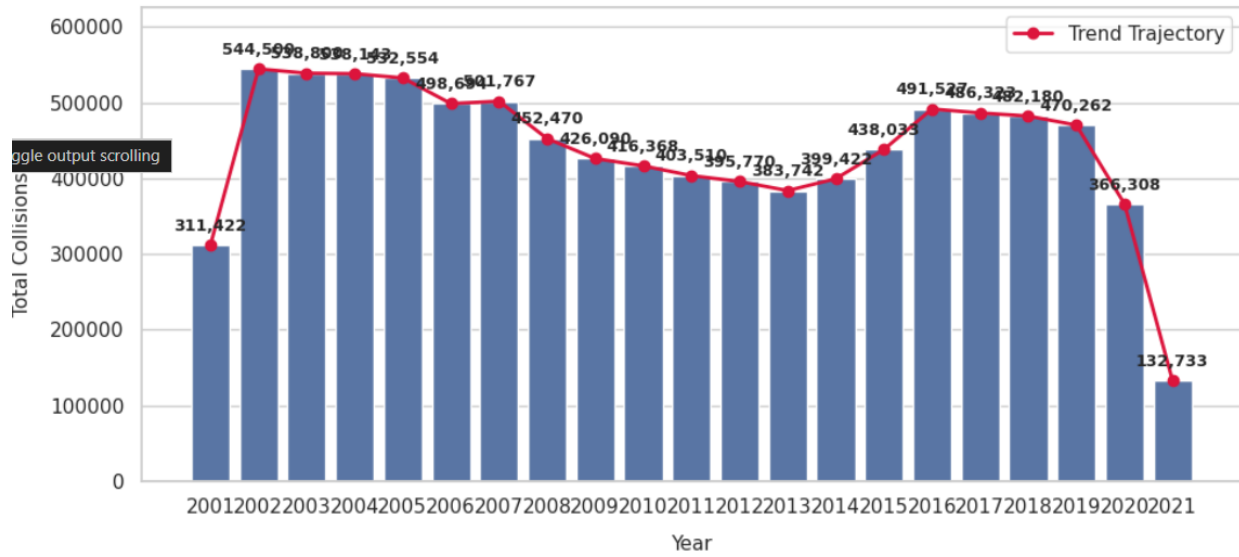


3.1.11. Collision Trends Over Time

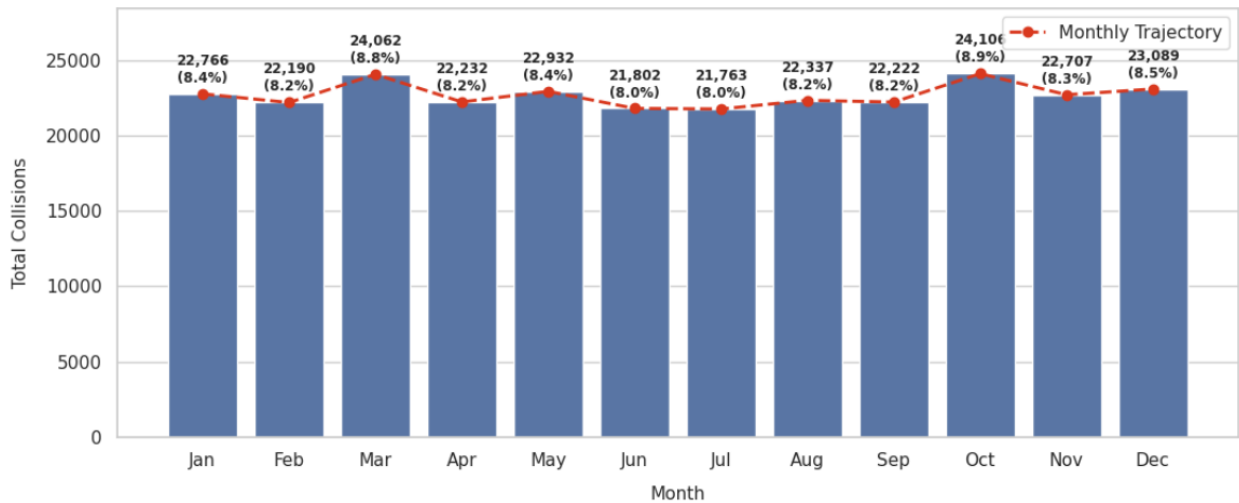
Extract and analyzing collision trends over time.

Q: Analyze yearly, monthly and hourly trends in collisions.

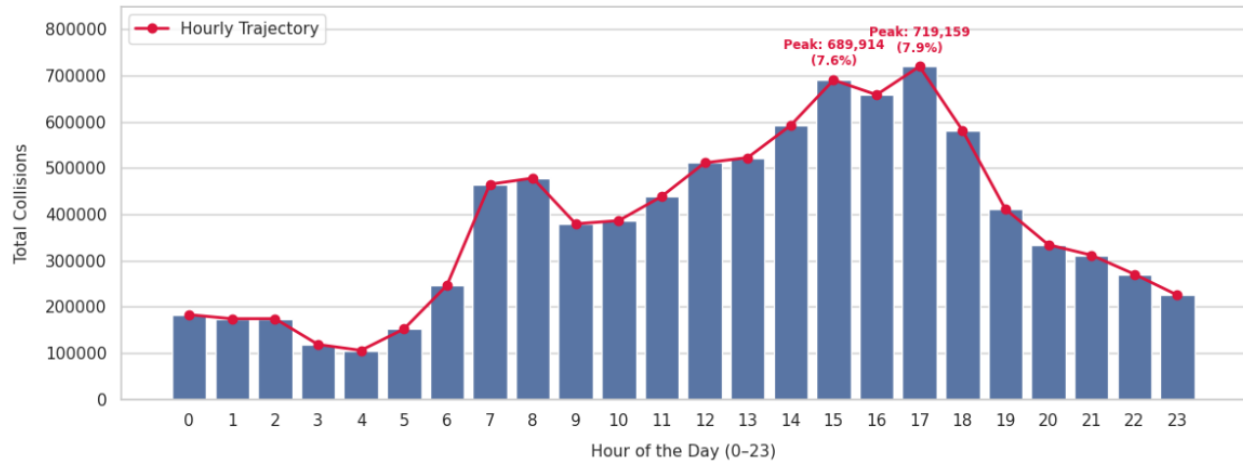
Yearly Collision Volume Trend



Aggregated Monthly Collision Trend (Seasonality)



Hourly Collision Volume Distribution (24-Hour Cycle)



4. ETL Querying

4.1. Top 5 Counties

Q: Identify the top 5 counties with the highest number of collisions.

```
# Query: Identify the top 5 counties with the most collisions
df_final.createOrReplaceTempView("collisions")

# PySpark SQL Query to find the top 5 counties with the most collisions
top_5_counties = spark.sql("""
    SELECT
        county_location,
        COUNT(*) AS collision_count
    FROM collisions
    WHERE county_location IS NOT NULL
      AND county_location != 'Unknown'
    GROUP BY county_location
    ORDER BY collision_count DESC
    LIMIT 5
""")

top_5_counties.show(truncate=False)
```

```
+-----+-----+
|county_location|collision_count|
+-----+-----+
|los angeles    |2800144        |
|orange         |716208         |
|san bernardino |556855         |
|san diego      |521565         |
|riverside      |481665         |
+-----+-----+
```

4.2. Month with Highest Collisions

Q. Identify the month with the highest number of collisions.


```
# Query: Find the month with the highest number of collisions
spark.sql("""
    SELECT
        MONTH(collision_date) AS month_num,
        DATE_FORMAT(collision_date, 'MMMM') AS month_name,
        COUNT(*) AS collision_count
    FROM collisions
    WHERE collision_date IS NOT NULL
    GROUP BY MONTH(collision_date), DATE_FORMAT(collision_date, 'MMMM')
    ORDER BY collision_count DESC
    LIMIT 1
""").show(truncate=False)
```

```
+-----+-----+-----+
|month_num|month_name|collision_count|
+-----+-----+-----+
|10       |October  |819610        |
+-----+-----+-----+
```

4.3. Weather Conditions with Highest Collisions.

Q. Determine the most common weather conditions during collisions.

```
# Query: Find the most common weather condition during collisions

most_common_weather = spark.sql("""
    SELECT
        weather_condition,
        COUNT(*) AS collision_count
    FROM collisions
    WHERE weather_condition IS NOT NULL
        AND weather_condition != 'Unknown'
    GROUP BY weather_condition
    ORDER BY collision_count DESC
    LIMIT 1
""")

most_common_weather.show(truncate=False)
```

```
+-----+-----+
|weather_condition|collision_count|
+-----+-----+
|clear            |7593103        |
+-----+-----+
```

4.4. Fatal Collisions.

Q. Calculate the percentage of collisions that resulted in fatalities.

```
# Query: Determine the percentage of collisions that resulted in fatalities
fatal_percentage_df = spark.sql("""
    SELECT
        COUNT(*) AS total_collisions,
        SUM(CASE WHEN killed_victims > 0 OR LOWER(severity) = 'fatal' THEN 1 ELSE 0 END) AS fatal_collisions,
        ROUND(
            (SUM(CASE WHEN killed_victims > 0 OR LOWER(severity) = 'fatal' THEN 1 ELSE 0 END) * 100.0) / COUNT(*),
            2
        ) AS fatal_percentage
    FROM collisions
""")

fatal_percentage_df.show(truncate=False)
```

total_collisions	fatal_collisions	fatal_percentage
9210618	64342	0.70

4.5. Dangerous Time for Collisions

Q. Find the most dangerous time of day for collisions.

```
# Query: Find the most dangerous time of day for collisions
dangerous_time_df = spark.sql("""
    SELECT
        collision_hour,
        COUNT(*) AS collision_count
    FROM collisions
    WHERE collision_hour IS NOT NULL
        AND collision_hour BETWEEN 0 AND 23
    GROUP BY collision_hour
    ORDER BY collision_count DESC
    LIMIT 1
""")

dangerous_time_df.show(truncate=False)
```

collision_hour	collision_count
17	719159

4.6. Road Surface Conditions

Q. Identify the top 5 road surface conditions with the highest collision frequency.

Query: List the top 5 road types with the highest collision frequency

```
top_5_road_surface = spark.sql("""
```

```
    SELECT
        road_surface,
        COUNT(*) AS collision_count
    FROM collisions
    WHERE road_surface IS NOT NULL
        AND road_surface != 'Unknown'
    GROUP BY road_surface
    ORDER BY collision_count DESC
    LIMIT 5
```

```
""")
```

```
top_5_road_surface.show(truncate=False)
```

```
+-----+-----+
|road_surface|collision_count|
+-----+-----+
|dry         |8328994        |
|wet         |750571         |
|snowy       |40133          |
|slippery    |10471          |
|H           |101            |
+-----+-----+
```

4.7. Lighting Conditions

Q. Analyze lighting conditions that contribute to the highest number of collisions.

```
# Query: Find the top 3 lighting conditions that lead to the most collisions
top_3_lighting = spark.sql("""
    SELECT
        lighting_condition,
        COUNT(*) AS collision_count
    FROM collisions
    WHERE lighting_condition IS NOT NULL
        AND lighting_condition != 'Unknown'
    GROUP BY lighting_condition
    ORDER BY collision_count DESC
    LIMIT 3
""")

top_3_lighting.show(truncate=False)
```

lighting_condition	collision_count
daylight	6133074
dark with street lights	1954725
dark with no street lights	729810

5. Conclusion

Write your conclusion.

Working through this 9.2-million-row SWITRS dataset in PySpark highlighted clear differences between what causes sheer collision volume versus what drives incident severity. From an ETL and pipeline design perspective, cleaning the raw records, dropping sparse columns with over 60% nulls, and saving optimized Parquet files gave us a solid foundation to query the numbers accurately.

Core Findings from the Data

- Exposure over Weather Extremes: Counterintuitively, over 82% of collisions occurred under clear skies, and more than 90% happened on dry roads. High accident totals are primarily a product of traffic density and congestion rather than adverse weather.
- Commute Pressure Points: Collision volume tracks commuter flow, climbing steadily across the workweek to peak on Friday (16.5%). Daily traffic spikes sharply at 17:00 (5:00 PM), which logged over 719,000 crashes statewide.

- **Geographic Bottlenecks:** Accidents cluster tightly in major urban hubs. Southern California carries most of the burden, with Los Angeles County alone accounting for over 30.4% (~2.8M) of all incidents, followed by Orange (7.8%) and San Bernardino (6.0%).
- **Infrastructure vs. Severity Risk:** While absolute numbers are highest during daylight, collisions on roads marked "dark with no street lights" yielded more than double the proportion of severe injuries (~4.25%) compared to daylight (~1.93%). Poor visibility directly compounds crash severity.
- **Demographic Vulnerability:** The age distribution peaks heavily around young motorists (ages 18 to 25), pointing to driver inexperience as a major recurring risk factor.

Practical Engineering and Policy Takeaways

- **Targeted Caltrans Investments:** State funding should prioritize lighting installations and safety barrier retrofits in high-density corridors across Los Angeles, Orange, and San Bernardino counties, targeting unlit arterial roads first.
- **Dynamic Traffic Management:** Transportation agencies can use the 16:00–19:00 weekday window to deploy adaptive signal timing and variable highway speed limits, dampening the shockwave braking that fuels high collision rates.
- **Vulnerable Road User Protection:** City planning boards need to prioritize physical barriers and dedicated signal phases at dense urban intersections rather than standard painted lanes.
- **Next Steps for the Data Pipeline:** With the cleaned tables partitioned and stored in Parquet/AWS S3, the logical progression is building an automated feature store. Joining live Caltrans sensor feeds with historical weather and crash density will let us train MLlib classification models to flag high-risk segments before the evening rush hour begins.

6. Visualization Integration using PowerBI

